

Solving Quadratic Equations by Factoring (Homework)

Solve the following quadratic equations.

① $x^2 + 2x - 15 = 0$

$x = -5 \text{ or } 3$

② $y^2 + 13y + 40 = 0$

$y = -8 \text{ or } -5$

③ $0 = b^2 - 9b - 10$

$b = 10 \text{ or } -1$

④ $0 = x^2 - 8x + 16$

$x = 4$

⑤ $y^2 - 3y + 2 = 0$

$y = 2 \text{ or } 1$

⑥ $b^2 - 10b - 24 = 0$

$b = 12 \text{ or } -2$

⑦ $0 = x^2 + 10x + 25$

$x = -5$

⑧ $y^2 + 14y - 15 = 0$

$y = -15 \text{ or } 1$

⑨ $0 = b^2 + 10b + 16$

$b = -8 \text{ or } -2$

⑩ $2x^2 + 9x - 35 = 0$

$x = -7 \text{ or } \frac{5}{2}$

⑪ $3y^2 + 32y + 20 = 0$

$y = -10 \text{ or } -\frac{2}{3}$

⑫ $4b^2 - 28b + 45 = 0$

$b = \frac{5}{2} \text{ or } \frac{9}{2}$

⑬ $0 = 5x^2 + 2x - 16$

$x = \frac{8}{5} \text{ or } -2$

⑭ $6y^2 - 5y + 1 = 0$

$y = \frac{1}{3} \text{ or } \frac{1}{2}$

⑮ $8b^2 + 11b + 3 = 0$

$b = -1 \text{ or } -\frac{3}{8}$

⑯ $12y^2 + 16y + 5 = 0$

$y = -\frac{5}{6} \text{ or } -\frac{1}{2}$

Write the following quadratic equations in standard form; then solve.

⑰ $-11x + x^2 + 24 = 0$

$x = 8 \text{ or } 3$

⑱ $y^2 - 12y = -35$

$y = 7 \text{ or } 5$

⑲ $b^2 = -b + 56$

$b = -8 \text{ or } 7$

⑳ $-x - 6 = -x^2$

$x = 3 \text{ or } -2$

㉑ $y^2 + 15y + 50 = 0$

$y = -10 \text{ or } -5$

㉒ $b^2 - 33 = 8b$

$b = 11 \text{ or } -3$

㉓ $0 = 2x^2 - 30 + 17x$

$x = \frac{3}{2} \text{ or } -10$

㉔ $-40 + 3y^2 + 19y = 0$

$y = -8 \text{ or } \frac{5}{3}$

㉕ $1 = 8b^2 - 2b$

$y = \frac{1}{2} \text{ or } -\frac{1}{4}$